

Shashank Karki

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Education

VIRGINIA POLYTECHNIC INSTITUTE AND STATE UNIVERSITY

Blacksburg, VA

PhD. in Geospatial and Environmental Analysis

2024-Present

Thesis: Enhancing Urban spaces with GeoAI and visual language models

Advisor: Dr. Robert Oliver and Dr. Junghwan Kim

Relevant Coursework: Machine Learning, Big Data Engineering, GenAI for Social Sciences

VIRGINIA POLYTECHNIC INSTITUTE AND STATE UNIVERSITY

Blacksburg, VA

Masters in Geography, GPA: 4.0

2022-2024

Thesis: Tracking Human Movement and Occupancy Indoors Using Terrestrial Lidar

Advisor: Dr. Thomas Pingel

Relevant Coursework: Programming for GIS, Enterprise GIS, Analysis of Spatial Data, Advanced Analysis in GIS, Advanced Topics in Remote Sensing

KATHMANDU UNIVERSITY

Dhulikhel, Nepal

Bachelor of Engineering in Geomatics Engineering, GPA: 3.84

2014-2018

Thesis: Analyzing the Effect of Image Overlap, Number of GCPs & Distribution Pattern on the Overall Accuracy of UAV Photogrammetry

Relevant Coursework: Python Programming for GIS, Photogrammetry, Spatial Database Management System, Remote Sensing, Geographic Information System

Experience

VIRGINIA POLYTECHNIC INSTITUTE AND STATE UNIVERSITY

Blacksburg, VA

Graduate Teaching Assistant, Department of Geography

Jan 2022 – Dec 2024

- Delivered instructional support for various GIS courses, including Principles of GIS, Modeling with GIS, and Programming for GIS, as well as Remote Sensing.
- Provided lab assistance and support for the lead instructor.
- Enhanced skills in teaching, verbal communication, critical reasoning, and leadership.

VIRGINIA POLYTECHNIC INSTITUTE AND STATE UNIVERSITY

Blacksburg, VA

Graduate Research Assistant, Department of Geography

Assessment of Accessibility and Transportation Network Modeling

June 2023 – August 2023

- Conducted comprehensive accessibility assessments for dental care services across Virginia.
- Managed and analyzed public transit data using General Transit Feed Specification (GTFS) files.
- Developed transportation network models utilizing Rapid Realistic Routing (R5R) in R.
- Employed Pandas and GeoPandas for extensive travel time data analysis and management.

Lidar Sensor Data Analysis and Object Detection

June 2023 – August 2023

- Managed and calibrated lidar sensors for optimal data collection.
- Analyzed lidar data to detect space use patterns and identify moving objects.
- Developed deep learning models for object detection in lidar feeds.
- Created and implemented accuracy metrics, including bidimensional regression, chamfer distance, and chamfer ratio, for method comparison.

Green Exposure Index Calculation and Image Segmentation

May 2023 – June 2023

- Calculated the Green Exposure Index for home to work commutes in Virginia.
- Utilized Google Directions API and Street View API for route calculation and imagery.
- Applied PixelLib for image segmentation and greenery ratio calculations.
- Performed data aggregation and analysis using Pandas.

VIRGINIA POLYTECHNIC INSTITUTE AND STATE UNIVERSITY **Graduate Research Assistant, Department of Fish and Wildlife Department**

Blacksburg, VA
May 2022 – Aug 2022

- Computed Land Surface Temperature data at a 30m resolution.
- Conducted an assessment of habitat changes in Bog Turtle populations from the 1980s-2020s.
- Utilized Google Earth Engine for satellite imagery analysis and management.
- Managed and performed spatial overlay operations on over 10 TBs of satellite imagery using High-Performance Computing (HPC) resources.

KATHMANDU UNIVERSITY **Graduate Teaching Assistant, Department of Geomatics Engineering**

Dhulikhel, Nepal
Mar 2019 – Oct 2021

- Conducted lectures and practical classes on GIS, Theory of Errors & Adjustments, and Control Survey.
- Led survey camps, instructing and managing a group of 60 students.
- Maintained and managed equipment in both the Survey and Photogrammetry Labs.

GEONODE NEPAL, FUNDED BY THE WORLD BANK GROUP **Spatial Database Manager**

Remote
Nov 2019 – Nov 2021

- Managed and updated a free spatial data repository for Nepal, available at GeoNode Nepal.
- Developed expertise in geospatial data management over geoportals through hands-on experience and research.

NEPAL ELECTRICITY AUTHORITY **UAV and DGPS Surveyor**

Kathmandu, Nepal
June 2019 – July 2021

- Conducted UAV surveys using Mavic 2 Pro for a prospective hydropower plant site in Dolpa, Nepal.
- Processed aerial imagery to generate orthomosaics, Digital Terrain Models (DTM), and Digital Surface Models (DSM) using Pix4D.
- Carried out Differential GPS Surveys to establish accurate ground control points for geospatial data correction.

- Estimated theoretical hydropower potential within specified regions.
 - Performed hydrological modeling to assess water flow and resource availability.
 - Utilized ArcHydro Tools in ArcMap for spatial analysis.
 - Integrated precipitation datasets from WorldClim for comprehensive environmental modeling.
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Research

PUBLICATIONS

- (In preparation) Wang, C., **Karki, S.**, & Zhang, M. (2026). Shifts in dental service areas following Medicaid policy change: A statewide analysis of Virginia Medicaid claims. (Expected submission: April 2026).
- (In preparation) Kim, J., **Karki, S.**, Gyanwali, S., Chandramouli, V., David-John, B., & Bloom, M. (2026). From presence to perception: Quantifying human engagement with urban greenery using an AI-powered wearable eye-tracking device. Expected submission: March 2026.
- (In preparation) **Karki, S.**, Kim, J., Hankey, S., Ye, X., & Oliver, R. (2026). Generative AI for urban and spatial planning: A systematic review of methods, data practices, and spatial reasoning. Expected submission: March 2026.
- (In review) **Karki, S.**, Kim, J., Oliver, R., & Jang, K. M. (2025). An AI-driven framework for enhancing urban walkability using image analysis, augmentation, and community-centric improvements. Manuscript in review.
- Sherman, Z., **Karki, S.**, Brickhouse, T., Vujicic, M., Nasseh, K., Wang, C., Kim, H., Kim, J., & Zhang, M. (2026). Geospatial access to oral health care in Virginia: Insights from driving and public transit analysis. *The Journal of the American Dental Association*, 157(1), 36-46.e6. <https://doi.org/10.1016/j.adaj.2025.09.003>
- Casado, G. L., **Karki, S.**, John, V., Kim, J., & Zhang, M. (2025). Transit-Based Model to Evaluate Geospatial Accessibility of Eye Care in Virginia. *Investigative Ophthalmology & Visual Science*, 66(8), 2100–2100.
- Flack, A. H., Pingel, T. J., Baird, T. D., **Karki, S.**, & Abaid, N. (2025). Lidar-Based Detection and Analysis of Serendipitous Collisions in Shared Indoor Spaces. *Remote Sensing*, 17(18), 3236. <https://doi.org/10.3390/rs17183236>
- Zhang, T., Yu, H., & **Karki, S.** (2025). Empowering entrepreneurs: age, telework, and geographic context in transitioning to knowledge-based self-employment. *Entrepreneurship & Regional Development*, 1-37.
- Kim, J., **Karki, S.**, Brickhouse, T., Vujicic, M., Nasseh, K., Wang, C., & Zhang, M. (2025). Navigating Disparities in Dental Health—A Transit-Based Investigation of Access to Dental Care in Virginia. *Community Dentistry and Oral Epidemiology*, 53(1), 117-124.
- **Karki, S.**, Pingel, T. J., Baird, T. D., Flack, A., & Ogle, T. (2024). Enhancing Digital Twins with Human Movement Data: A Comparative Study of Lidar-Based Tracking Methods. *Remote Sensing*, 16(18), 3453.
- Gyanwali, S., **Karki, S.**, Jang, K. M., Crawford, T., Zhang, M., & Kim, J. (2024). Implications for spatial non-stationarity and the neighborhood effect averaging problem (NEAP) in green inequality research: evidence from three states in the USA. *Journal of Geographical Systems*, 1-24.
- Bista, N., Mahat, D., Manandhar, S., Regmi, B., Panday, U. S., & **Karki, S.** (2021). Analyzing Trend and Pattern

of Agricultural Drought: A Case Study of Karnali and Sudurpashchim Provinces. *Journal on Geoinformatics, Nepal*, 20(1), 1–8. <https://doi.org/10.3126/njg.v20i1.39470>

- Shrestha, A., Ghimire, S., Pokharel, A., Dhimi, S., **Karki, S.**, Awasthi, B. and Acharya, T., (2021). Landslide susceptibility mapping in Budhi Gandaki River subbasin using frequency ratio and statistical Information index. *Bulletin of Nepal Hydrogeological Association*, 6.
- Awasthi B., **Karki S.**, Regmi P., Dhimi D.S., Thapa S., Panday U.S. (2020) Analyzing the Effect of Distribution Pattern and Number of GCPs on Overall Accuracy of UAV Photogrammetric Results. In: Jain K., Khoshelham K., Zhu X., Tiwari A. (eds) *Proceedings of UASG 2019*. UASG 2019. Lecture Notes in Civil Engineering, vol 51. Springer, Cham. https://doi.org/10.1007/978-3-030-37393-1_29
- Shrestha, R., Zevenbergen, J., Panday, U. S., Awasthi, B., and **Karki, S.**, (2019). Revisiting the Current UAV Regulations in Nepal: A Step towards Legal Dimension for UAV's Efficient Application, *Int. Arch. Photogramm. Remote Sens. Spatial Inf. Sci.*, XLII-5/W3, 107–114, <https://doi.org/10.5194/isprs-archives-XLII-5-W3-107-2019>
- Awasthi, B., **Karki, S.**, Regmi, P., Dhimi, D. S., Thapa, S., & Panday, U. S. (2019). Assessing The Overall Accuracy of UAV Photogrammetry: A Methodology for Different Combination of Imagery Parameters and Terrain Types. *Journal of Land Management and Geomatics Education*, 01(1), ISSN: 2676-1408, 22–32.

PRESENTATIONS

- **Karki, S.**, Kim, J., Oliver, R., & Jang, K. M. (2025). GeoAI twins for walkable cities: Modeling urban perception with generative image analysis. Presented at the 2025 CGA Conference: The Geography of Digital Twins, Cambridge, Massachusetts, May 22–24.
- **Karki, S.**, Kim, J., & Oliver, R. (2025). An AI-driven framework for enhancing urban walkability using image analysis, augmentation, and community-centric improvements. American Association of Geographers (AAG) Annual Meeting, Detroit, MI. , March 24–28.
- **Karki, S.**, Pingel, T.J., Flack, A., & Baird, T.D. (2023). Aggregating Multi-Sensor Terrestrial Lidar Data for Analyzing Human Behavior. Presented at the 2023 ASPRS International Technical Symposium, June 12-16.

INVITED TALKS

- **Karki, S.** (2023). Capturing and Modelling Movement and Behavior in a Digital Twin. SPARC Workshop on Digital Twins in Geospatial Research, Arizona State University, February 27–28.
- **Karki, S.** (2025). An AI-driven framework for enhancing urban walkability using image analysis, augmentation, and community-centric improvements. GEOG 4304/5304: Geospatial Analysis of Mobility, Virginia Tech. November

POSTERS

- **Karki, S.**, Casado, G. L., John, V. J., Kim, J., & Zhang, M. (2025, May). Eyes on transit: Disparities in accessing eye care by public transportation in Virginia [Poster presentation]. Carilion Research Day, Virginia Tech Carilion School of Medicine, Roanoke, VA.
- **Karki, S.**, Jang, K. M., Oliver, R., & Kim, J. (2025, April). Enhancing urban walkability with AI: Image processing, augmentation, and community-driven solutions. OGIS Symposium 2025, Office of GIS and Remote Sensing, Virginia Tech
- **Karki, S.**, Flack, A., Pingel, T.J., and Baird, T.D. (2023). Navigating the Indoor Frontier: Uncovering Movement and Occupancy Patterns with Terrestrial Lidar. Virginia Tech OGIS Symposium. April 21.
- **Karki, S.**, Wilson, B., Granger, A., Flack, A., Neal, C., Dressel, L., Carruthers, C., and Pingel, T. (2022). Indoor

Mapping. Virginia Tech OGIS Symposium. April 9.

- **Karki, S.,** and, Rijal, S. (2022). Linear Spectral Unmixing for Vegetation Estimates at Different Spatial Resolutions. Virginia Tech OGIS Symposium. April 9.
- **Karki, S.,** (2019). Non-Governmental GeoNode: A geospatial content management system for Nepal. Presented at the International Workshop on "Capacity Building and Education Outreach in Advanced Geospatial Technologies and Land Management", LMTC, Dhulikhel, Kavre, December 10-11.

Leadership and Organizational Experience

FOSS4G ASIA 2021

Kathmandu, Nepal

Local Organizing Committee

February 2021

- Led the planning and execution of a conference schedule and venue management managing over 150 attendees.
- Coordinated with different departments, secured sponsorships, and managed a team of volunteers.
- Led a team of about 20 volunteers

INTERNATIONAL GIS DAY 2017, KATHMANDU UNIVERSITY

Kathmandu, Nepal

Event Coordinator

March 2017

- Led the development and preparation of the event program, ensuring a comprehensive and engaging schedule for attendees.
- Acted as the official host for the event, facilitating smooth transitions between segments and maintaining audience engagement.

Honors and Awards

INDUCTION INTO THE HONOR SOCIETY OF PHI KAPPA PHI

Blacksburg, VA

Virginia Polytechnic Institute & State University

April 2023

- Recognition: Outstanding academic excellence
- Selection: By invitation, requiring nomination and chapter approval
- Eligibility: Top 10% of seniors and 7.5% of juniors, along with eligible graduate students, faculty, professional staff, and distinguished alumni

BEST PAPER AWARD

Roorkee, India

Indian Institute of Technology

April 2019

- Event: International Conference on Unmanned Aerial System in Geomatics 2019
- Date: April 06-07, 2019
- Location: IIT Roorkee
- Recognition: Outstanding contribution and presentation of paper

Academic Excellence Scholarship

Dhulikhel, Nepal

Kathmandu University

April 2019

- Institution: Department of Geomatics Engineering, Kathmandu University
- Semesters: 1, 3, 5, 6 and 8
- Recognition: Awarded for achieving the highest academic performance in the respective semesters during undergraduate studies

Skills & Interests

TECHNICAL

- Geographic Information Systems (GIS):
 - Proficient: ArcGIS Suite (ArcMap, ArcGIS Pro, ArcGIS Online, ArcGIS Enterprise), QGIS, ERDAS IMAGINE, Google Earth, Google Earth Engine
 - Familiar: Leaflet, GeoDa, ArcGIS Server, AutoCAD
- Photogrammetry:
 - Pix4D, WebODM, Agisoft Metashape
- Programming Languages:
 - Intermediate/Working Proficiency: Python, R
 - Beginner: C, C++, Java, JavaScript
 - Python Libraries: Numpy, Pandas, GeoPandas, Scipy
 - Development Environments: Spyder, Jupyter Notebook, JupyterLab, Google Colab
- Database Management:
 - Sqlite, PostGIS, MySQL, PostgreSQL, Apache Spark
- Operating Systems:
 - Proficient: Linux (Ubuntu)
- Web Mapping and Geospatial Web Development:
 - Mapbox, OpenLayers, Geoserver, PostGIS Web Services
- Version Control:
 - Git (GitHub)
- Cloud Platforms:
 - Amazon Web Services
- Other Skills:
 - Microsoft Office Suite

LANGUAGES

- English (Fluent)
 - Nepali (Native)
 - Hindi (Intermediate)
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